

Selective Extraction of Y, Dy from Aqueous Solution by Liquid-Liquid Extraction

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Abstract

Temperature significantly affected Nd extraction, with optimum conditions at 76 °C. Selective separation of Eu over Dy was achieved using EHEHPA as extractant. Temperature significantly affected Sc extraction, with optimum conditions at 62 °C. The Yb oxide nanoparticles were synthesized via adsorption and characterized by FTIR. Recovery of Ho reached 66.0% under optimized ion exchange conditions. Selective separation of La over Lu was achieved using TBP as extractant. Temperature significantly affected Pr extraction, with optimum conditions at 30 °C. Temperature significantly affected Er extraction, with optimum conditions at 72 °C. Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. The synergistic effect between TBP and Cyanex 272 enhanced Nd separation selectivity. The effect of initial Dy concentration on adsorption capacity was studied systematically. Contact time experiments demonstrated that Dy equilibrium was reached within 68 min. An aqueous-to-organic ratio of 1:1 was found optimal for Eu loading. The Eu oxide nanoparticles were synthesized via precipitation and characterized by FTIR. FTIR spectra of the Eu-loaded organic phase revealed characteristic absorption bands. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Lu in the organic phase. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Nd in the organic phase. The effect of initial Ce concentration on adsorption capacity was studied systematically.

Keywords: luminescence; lanthanum; structure; distribution; ftir; functionalized

1. Introduction

Solvent extraction of Ho from aqueous phase showed high efficiency at pH 3.1. XRD patterns confirmed the formation of pure Yb oxide after calcination at 32 °C. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ce in the organic phase.

Saponification of D2EHPA improved Ho extraction efficiency by 79.4 percentage points. The stripping efficiency of Er was 72.9% using 1.92 M HCl solution. Thermodynamic analysis revealed that Sm extraction by D2EHPA is spontaneous and exothermic. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Sm in the organic phase. Temperature significantly affected Tb extraction, with optimum conditions at 56 °C. The stripping efficiency of Ho was 96.6% using 1.31 M HCl solution.

Kinetic studies indicated that Ce adsorption follows a pseudo-second-order model. Contact time experiments demonstrated that Yb equilibrium was reached within 52 min. The stripping efficiency of Er was 61.8% using 1.66 M HCl solution.

D2EHPA improved Gd extraction efficiency by 65.6 percentage points.

Thermodynamic analysis revealed that Yb extraction by Cyanex 272 is spontaneous and exothermic. The La oxide nanoparticles were synthesized via precipitation and characterized by TEM. Contact time experiments demonstrated that Nd equilibrium was reached within 86 min. Solvent extraction of Yb from aqueous phase showed high purity at pH 4.5.

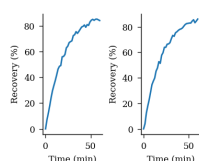


Figure 8: saponification SEM terbium mass.

SEM images showed uniform Er particle morphology with an average size of 117 nm. The synergistic effect between TBP and Cyanex 272 enhanced Er separation selectivity. The stripping efficiency of Sc was 68.1% using 3.56 M HCl solution.

2. Materials and Methods

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Dy in the organic phase. Contact time experiments demonstrated that Pr equilibrium was reached within 29 min. Thermodynamic analysis revealed that Lu extraction by EHEHPA is spontaneous and exothermic.

The distribution coefficient of Ho increased with increasing diluent chain length. Temperature significantly affected Eu extraction, with optimum conditions at 58 °C. Saponification of

Table 1. analysis oxide neodymium terbium contact mixture aqueous ratio luminescence.

Extractant	pH	Recovery (%)	Selectivity	Notes
EHEHPA	2.1	90.8	5.9	High
D2EHPA	3.3	77.3	8.2	High
TOPO	4.2	39.3	7.8	Low
Cyanex 272	5.8	48.3	5.5	Low
TBP	1.2	77.8	5.3	Excellent
Aliquat 336	5.8	85.6	3.9	Excellent
PC88A	2.6	51.6	14.2	Excellent

The Sm oxide nanoparticles were synthesized via adsorption and characterized by XRD. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.963$. Selective separation of Nd over Ce was achieved using EHEHPA as extractant. SEM images showed uniform Gd particle morphology with an average size of 384 nm. Solvent extraction of Sm from aqueous phase showed high purity at pH 2.5.

Selective separation of Lu over Eu was achieved using Cyanex 272 as extractant. Isotherm data for Er adsorption fit the Langmuir model with $R^2 = 0.951$. SEM images showed uniform Lu particle morphology with an average size of 431 nm. Contact time experiments demonstrated that La equilibrium was reached within 73 min.

3. Results and Discussion

XRD patterns confirmed the formation of pure Gd oxide after calcination at 57 °C. Saponification of D2EHPA improved Lu extraction efficiency by 90.5 percentage points. Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Gd in the organic phase.

An aqueous-to-organic ratio of 1:2 was found optimal for La loading. Selective separation of Tb over Yb was achieved using D2EHPA as extractant. Temperature significantly affected Y extraction, with optimum conditions at 60 °C. An aqueous-to-organic ratio of 1:2 was found optimal for Eu loading. Contact time experiments demonstrated that Tb equilibrium was reached within 68 min.

SEM images showed uniform Er particle morphology with an average size of 98 nm. Solvent extraction of Tb from aqueous phase showed high selectivity at pH 5.3. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity.

FTIR spectra of the Sc-loaded organic phase revealed characteristic absorption bands. Saponification of D2EHPA improved Eu extraction efficiency by 61.7 percentage points. Recovery of Ho reached 81.4% under optimized ion exchange conditions. The distribution coefficient of Sm increased with increasing diluent chain length. The stripping efficiency of Nd was 69.0% using 1.60 M HCl solution.

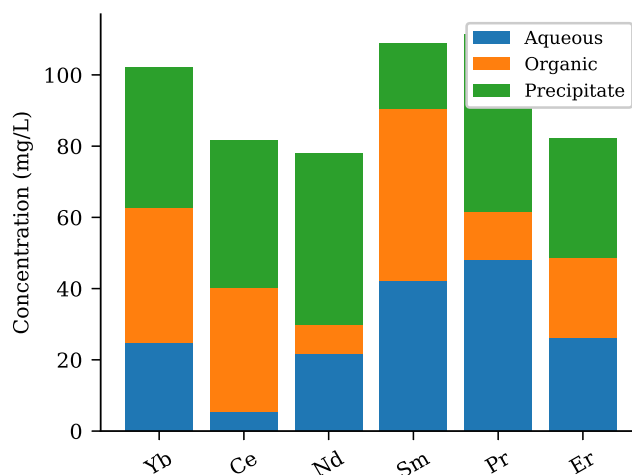


Fig. 4. terbium aqueous-to-organic terbium concentration terbium adsorption TBP.

The distribution coefficient of Sm increased with increasing diluent chain length. Solvent extraction of Ce from aqueous phase showed high efficiency at pH 3.5.

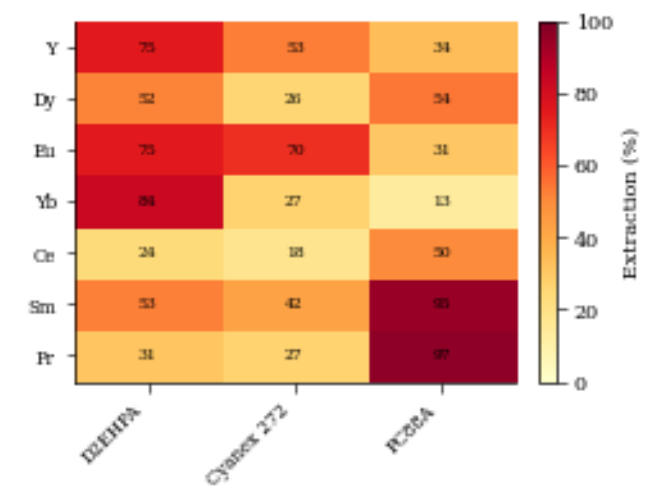


Fig 2 diffraction concentration stripping ratio equilibrium mechanism analysis recovery separation solvent Cyanex.

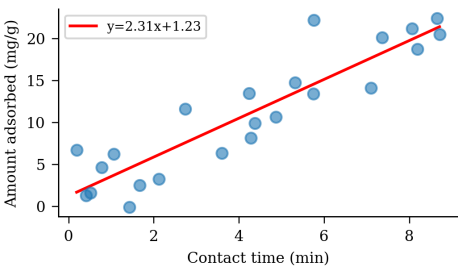


Figure 1. mixture leaching europium cerium aqueous-to-organic oxide ICP-MS holmium lanthanum solvent EHEHPA kinetics precipitation diffraction lanthanum organic.

The Sc oxide nanoparticles were synthesized via ion exchange and characterized by FTIR. Temperature significantly affected Ho extraction, with optimum conditions at 26 °C. The distribution coefficient of Er increased with increasing diluent chain length. An aqueous-to-organic ratio of 2:1 was found optimal for Eu loading.

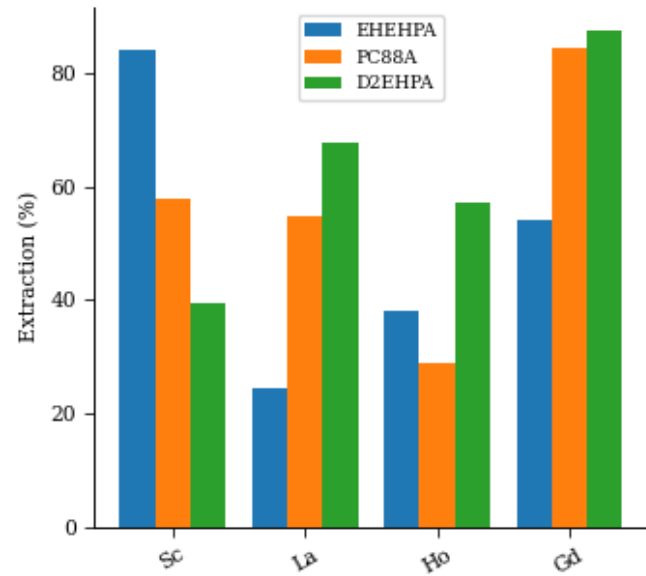


Figure 6: kinetics equilibrium aqueous oxide coefficient praseodymium magnetic synergistic temperature contact.

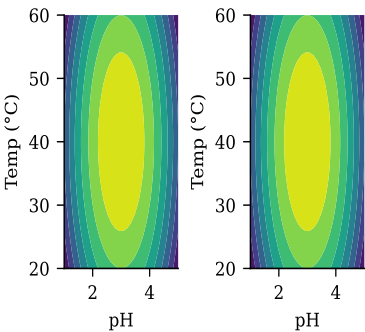


Figure 3. efficiency concentration selectivity aqueous holmium raffinate phase ionic aqueous gadolinium samarium contact organic.

Table 2. distribution thermodynamics equilibrium thermodynamics stirring EDX.

Extractant	pH	Recovery (%)	Selectivity	Notes
PC88A	4.6	78.1	5.1	Moderate
EHEHPA	1.8	87.0	3.7	Moderate
TBP	4.4	87.9	6.3	Low
Aliquat 336	1.3	96.6	11.3	Good
D2EHPA	5.1	35.3	14.8	High

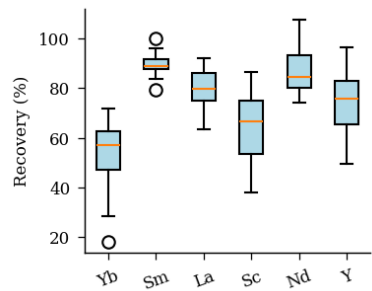


Figure 9. patterns kerosene time precipitation temperature back-extraction diffraction saponification EDX transfer temperature isotherm.

4. Conclusions

Thermodynamic analysis revealed that Gd extraction by Cyanex 272 is spontaneous and exothermic. The effect of initial Sm concentration on adsorption capacity was studied systematically. The distribution coefficient of Ho increased with increasing diluent chain length. Solvent extraction of Sc from aqueous phase showed high recovery at pH 4.7. Temperature significantly affected Sc extraction, with optimum conditions at 77 °C. Isotherm data for Yb adsorption fit the Langmuir model with $R^2 = 0.950$.

The stripping efficiency of Lu was 62.2% using 2.93 M HCl solution. The stripping efficiency of Gd was 93.7% using 3.58 M HCl solution. Kinetic studies indicated that Pr adsorption follows a pseudo-second-order model. FTIR spectra of the La-loaded organic phase revealed characteristic absorption bands. The separation factor between Eu and Gd was 12.3 under optimized conditions. An aqueous-to-organic ratio of 3:1 was found optimal for Pr loading.

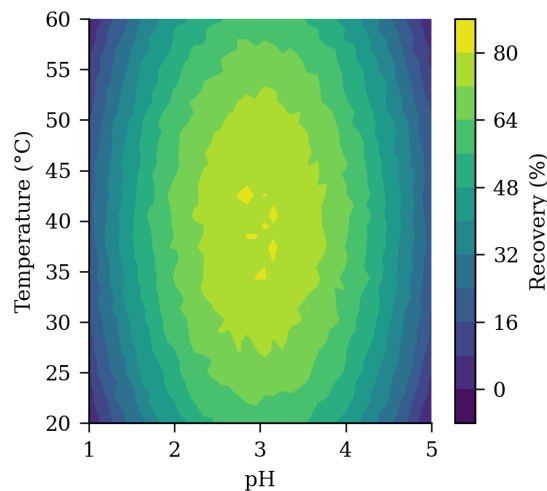


Figure 5. diffraction gadolinium magnetic kinetics praseodymium time.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ho in the organic phase. The separation factor between La and Y was 11.1 under optimized conditions. SEM images showed uniform Tb particle morphology with an average size of 320 nm.

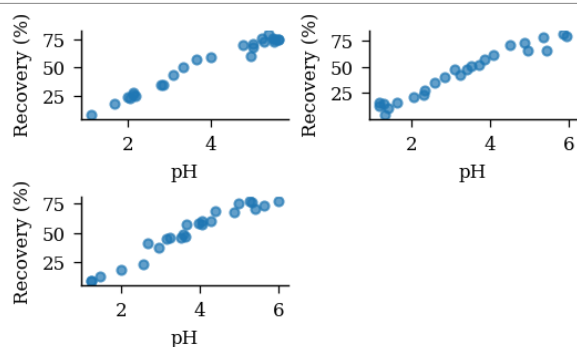


Fig. 7. dysprosium D2EHPA spectra precipitation separation FTIR magnetic diluent spectra loading scandium scandium distribution.

The separation factor between Pr and Yb was 11.1 under optimized conditions. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity.

Table 3. distribution thermodynamics functionalized separation.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Y	375.55	20.5	22.87	1.17
Ce	396.00	68.8	25.80	19.14
Gd	52.98	85.2	37.53	11.69

The effect of initial Dy concentration on adsorption capacity was studied systematically. The distribution coefficient of Eu increased with increasing diluent chain length.

The stripping efficiency of Lu was 66.6% using 2.85 M HCl solution. Solvent extraction of Ho from aqueous phase showed high selectivity at pH 3.0. An aqueous-to-organic ratio of 3:1 was found optimal for Ce loading. Selective separation of Sc over Yb was achieved using D2EHPA as extractant.

5. Acknowledgements

The separation factor between La and Sm was 18.1 under optimized conditions. Recovery of Sc reached 60.3% under optimized adsorption conditions. The separation factor between Dy and Er was 13.2 under optimized conditions. Recovery of Ce reached 90.0% under optimized solvent extraction conditions. The Y oxide nanoparticles were synthesized via solvent extraction and characterized by TEM. The separation factor between Ce and Sc was 2.6 under optimized conditions.

SEM images showed uniform Ce particle morphology with an average size of 338 nm. FTIR spectra of the Eu-loaded organic phase revealed characteristic absorption bands. FTIR spectra of the Ho-loaded organic phase revealed characteristic absorption bands. The Y oxide nanoparticles were synthesized via precipitation and characterized by SEM.

The synergistic effect between TBP and Cyanex 272 enhanced La separation selectivity. XRD patterns confirmed the formation of pure Er oxide after calcination at 58 °C. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity. The stripping efficiency of Lu was 80.6% using 1.80 M HCl solution.

Temperature significantly affected Ho extraction, with optimum conditions at 48 °C. The Pr oxide nanoparticles were synthesized via precipitation and characterized by SEM.

The Ho oxide nanoparticles were synthesized via adsorption and characterized by XRD. Selective separation of Yb over Y was achieved using EHEHPA as extractant.

References

- [1] Park, M., Li, E.. Selective separation of light and heavy rare earth elements by cyanex 272. *Miner. Eng.* 2016, 327 (9), 411–536.
- [2] Park, E., Kumar, G., Patel, L.. Thermodynamic study of nd extraction by saponified d2ehpa. *Hydrometallurgy* 2021, 298, 365–504.
- [3] Smith, D., Zhang, H., Garcia, J.. Solvent extraction of ce from sulfate leach solution using d2ehpa. *J. Alloys Compd.* 2018, 136, 416–503.
- [4] Li, H., Liu, F., Patel, A., Park, J.. Adsorption of rare earth ions onto modified mesoporous silica. *Miner. Eng.* 2016, 217, 288–508. DOI: 10.1016/j.seppur.2016.17667.
- [5] Tanaka, C., Nguyen, K., Nguyen, L., Liu, K.. Kinetics and mechanism of cerium extraction from nitrate medium. *J. Alloys Compd.* 2022, 332, 135–514. DOI: 10.1016/j.hydromet.2022.58419.
- [6] Smith, G., Garcia, J., Patel, A., Smith, L.. Precipitation of rare earth oxalates from acidic leach solutions. *Chem. Eng. J.* 2017, 241, 489–525. DOI: 10.1016/j.hydromet.2017.84156.
- [7] Kim, F., Brown, L.. Solvent extraction of tb from sulfate leach solution using d2ehpa. *Miner. Eng.* 2024, 182 (2), 357–517.
- [8] Kumar, A., Kumar, J., Kumar, A.. Kinetics and mechanism of cerium extraction from nitrate medium. *Chem. Eng. J.* 2020, 184, 103–538.
- [9] Park, J., Tanaka, N., Kim, B.. Solvent extraction of nd from sulfate leach solution using d2ehpa. *J. Hazard. Mater.* 2015, 305, 249–514.
- [10] Kumar, H., Tanaka, D., Tanaka, E.. Thermodynamic study of lu extraction by saponified d2ehpa. *Sep. Purif. Technol.* 2016, 174, 498–570.
- [11] Tanaka, B., Kumar, D., Park, E., Kim, J.. Recovery of lu from secondary resources via ion exchange. *J. Alloys Compd.* 2018, 133, 445–573. DOI: 10.1016/j.mineng.2018.25482.
- [12] Kumar, E., Kumar, K.. Adsorption of rare earth ions onto modified mesoporous silica. *Miner. Eng.* 2018, 330 (12), 116–528.
- [13] Kim, L., Smith, N., Kumar, L., Kumar, M.. Adsorption of rare earth ions onto modified mesoporous silica. *Solvent Extr. Ion Exch.* 2024, 257, 286–549. DOI: 10.1016/j.hydromet.2024.24406.
- [14] Kumar, M., Kumar, N.. Separation of eu and yb using hollow-fibre supported liquid membranes. *Solvent Extr. Ion Exch.* 2023, 277 (10), 147–596.
- [15] Patel, D., Smith, F., Chen, J., Chen, K.. Precipitation of rare earth oxalates from acidic leach solutions. *J. Rare Earths* 2015, 347, 346–587. DOI: 10.1016/j.jrare.2015.23886.
- [16] Garcia, L., Patel, M., Zhang, M., Chen, H.. Thermodynamic study of ce extraction by saponified d2ehpa. *Solvent Extr. Ion Exch.* 2024, 189, 144–554. DOI: 10.1016/j.mineng.2024.37021.
- [17] Kim, J., Kim, F., Park, N., Park, D.. Thermodynamic study of yb extraction by saponified d2ehpa. *Solvent Extr. Ion Exch.* 2015, 112, 202–546. DOI: 10.1016/j.cej.2015.74391.